

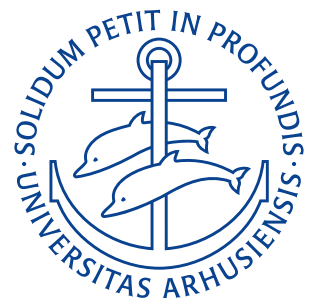
Department of Mechanical and Production Engineering

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Assignment 1

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Problem 1

Water flows steadily through a fire hose and nozzle. The hose is 35 mm in diameter and the nozzle tip is 25 mm in diameter; water gauge pressure in the hose is 510 kPa, and the stream leaving the nozzle is uniform. The exit speed and pressure are 32 m/s and atmospheric, respectively.

Find the force transmitted by the coupling between the nozzle and hose. Indicate whether the coupling is in tension or compression. State the assumptions made and sketch the situation.

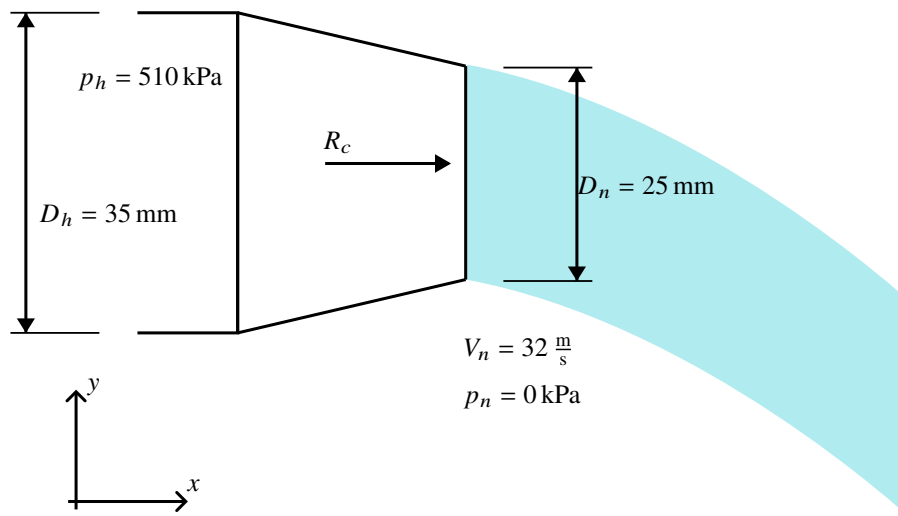


Figure 1.1: Sketch of the coupling.

Assumptions

- Steady flow
- Uniform flow at interfaces between coupling, hose, and nozzle
- Incompressible and frictionless liquid with negligible mass

Derivation

The momentum equation for an inertial control volume (as is the case for the coupling) is

$$\mathbf{F} = \mathbf{F}_B + \mathbf{F}_S = \frac{\partial}{\partial t} \int_{CV} \mathbf{V} \rho \, dV + \int_{CS} \mathbf{V} \rho \mathbf{V} \cdot d\mathbf{A} \quad (1.1)$$

In this case the flow is steady, hence $\frac{\partial}{\partial t} = 0$, hence the $\mathbf{F}_B = 0$. Reducing the vector equation to a scalar equation in the x -direction (defined on [Figure 1.1](#)) we obtain

$$F_x = F_{S_x} = \int_{CS} u \rho \mathbf{V} \cdot d\mathbf{A}.$$

As the flow is assumed uniform at both the inlet and outlet the integral further reduces to a sum as

$$F_x = F_{S_x} = \sum_{CS} u \rho \mathbf{V} \cdot \mathbf{A} \quad (1.2)$$

The forces in the x -direction on the control volume are the pressure at the interface between the hose and the coupling ($+p_h A_h$), the pressure at the nozzle ($-p_n A_n$), and the reaction on the coupling R_x . From Equation (1.2) we get

$$R_x + p_h A_h - p_n A_n = \rho Q (V_n - V_h) \quad (1.3)$$

As we are using gauge pressures $p_n = 0 \text{ kPa} \implies p_n A_n = 0 \text{ N}$, therefore Equation (1.3) reduces to

$$R_x + p_h A_h = \rho Q (V_n - V_h) \quad (1.4)$$

The continuity equation prescribes that the flow rate $Q = VA = \text{const.}$ Hence,

$$Q = A_n V_n \quad (1.5a)$$

$$V_h = \frac{Q}{A_h} = \frac{A_n V_n}{A_h} \quad (1.5b)$$

Substituting Equations 1.5 into Equation (1.4) and solving for R_x we obtain

$$\begin{aligned} R_x + p_h A_h &= \rho A_n V_n \left(V_n - \frac{A_n V_n}{A_h} \right) \\ R_x &= \rho A_n V_n^2 \left(1 - \frac{A_n}{A_h} \right) - p_h A_h \\ R_x &= 1000 \frac{\text{kg}}{\text{m}^3} \cdot \left(\pi \cdot (12,5 \text{ mm})^2 \right) \cdot \left(32 \frac{\text{m}}{\text{s}} \right)^2 \left(1 - \frac{(12,5 \text{ mm})^2}{(17,5 \text{ mm})^2} \right) - 510 \text{ kPa} \cdot \pi \cdot (17,5 \text{ mm})^2 \\ R_x &= -244,48 \text{ N}. \end{aligned}$$

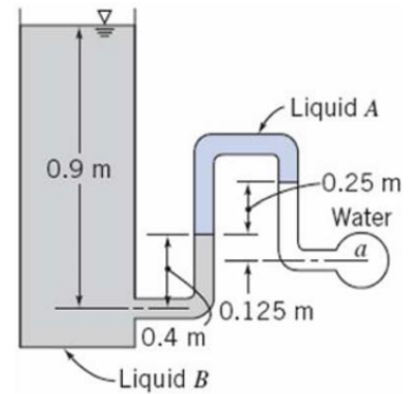
The negative sign (with $+x$ being to the right as shown on Figure 1.1) means the coupling reaction force points to the left, and the joint will therefore be in a tension of about 245 N.

$R_c = 245 \text{ N (Tension)}$

RESULT

Problem 2

Determine the gauge pressure in kPa at point a , if liquid A has $SG = 1,20$ and liquid B has $SG = 0,75$. The liquid surrounding point a is water, and the tank on the left is open to the atmosphere.



Assumptions

- All fluids are at rest
- All liquids are incompressible
- Temperature is uniform
- The left tank is open to the atmosphere, i.e. $p_b = 0$ gauge at the surface

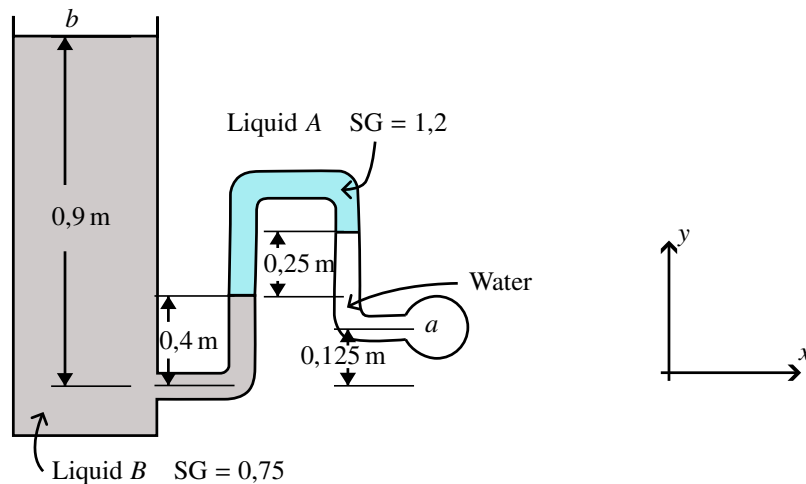


Figure 2.1: Sketch of the system

Derivation

For an incompressible fluid with density ρ the pressure p scales linearly with the depth h as

$$\Delta p = \rho g h.$$

Given two points separated by a series of fluids with densities and heights of ρ_i and h_i , respectively, the total change in pressure Δp is simply

$$\Delta p = g \sum_i \rho_i h_i \quad (2.1)$$

Starting at point a we go a distance h_1 “up” in the water corresponding to a height of

$$h_1 = 0,25 \text{ m (up to the } A \text{ /water-interface)} + 0,125 \text{ m} = 0,375 \text{ m}.$$

This corresponds to a pressure difference of

$$p_1 = p_a - \rho_w g h_1.$$

Inside liquid A a distance of $h_2 = 0,25 \text{ m}$ must be traversed vertically down to reach the A/B -interface. This gives a pressure difference of

$$p_2 = p_1 + \rho_A g h_2.$$

Now in liquid B the net rise to the open tank surface is $h_3 = 0,9 \text{ m} - 0,4 \text{ m} = 0,5 \text{ m}$. Which gives our atmospheric (gauge) pressure

$$p_{\text{atm}} = p_b = p_2 - \rho_B g h_3 = 0.$$

Combining these per [Equation \(2.1\)](#) yields

$$\begin{aligned} p_a - \rho_w g h_1 + \rho_A g h_2 - \rho_B g h_3 &= 0 \\ \implies p_a &= \rho_w g (h_1 - SG_A h_2 + SG_B h_3) \\ &= 1000 \frac{\text{kg}}{\text{m}^3} \cdot 9,81 \frac{\text{m}}{\text{s}^2} (0,375 \text{ m} - 1,2 \cdot 0,25 \text{ m} + 0,75 \cdot 0,5 \text{ m}) \\ &= 4,41 \text{ kPa}. \end{aligned}$$

The gauge pressure at a is therefore $p_a = 4,41 \text{ kPa}$.

Therefore a gauge pressure of $p_a = 4,41 \text{ kPa}$ can be measured at point a .

$p_a = 4,41 \text{ kPa (gauge)}$

RESULT

Problem 3

Determine the stream function ψ that will yield the velocity field.

$$\mathbf{V} = 2y(2x + 1)\hat{i} + (x(x + 1) - 2y^2)\hat{j} \text{ V.}$$

Assumptions

- Incompressible flow in the xy -plane

Derivation

The stream function for two-dimensional incompressible flow is defined as

$$u \equiv \frac{\partial \psi}{\partial y} \quad \text{and} \quad v \equiv -\frac{\partial \psi}{\partial x} \quad (3.1)$$

Starting with the u -coordinate we can integrate with respect to y as

$$\begin{aligned} \frac{\partial \psi}{\partial y} &= 2y + (2x + 1) \\ \psi(x, y) &= \int 2y(2x + 1) dy \\ &= 2xy^2 + y^2 + f(x). \end{aligned}$$

Doing the same with the v -coordinate but integrating with respect to x we obtain

$$\begin{aligned} -\frac{\partial \psi}{\partial x} &= x \cdot (x + 1) - 2y^2 \\ \psi(x, y) &= -\int (x \cdot (x + 1) - 2y^2) dx \\ &= -\frac{x^3}{3} - \frac{x^2}{2} + 2xy^2 + g(y). \end{aligned}$$

Comparing these we see that $f(x) = -\frac{x^3}{3} - \frac{x^2}{2}$ and $g(y) = y^2$. Hence the stream function is

$$\psi(x, y) = y^2 + 2xy^2 - \frac{x^2}{2} - \frac{x^3}{3}.$$

This solution can be verified by checking that the found stream function does not break the definition of the stream function presented in [Equation \(3.1\)](#) as

$$\begin{aligned} u(x, y) &= \frac{\partial}{\partial y} \left(y^2 + 2xy^2 - \frac{x^3}{3} - \frac{x^2}{2} \right) \implies u(x, y) = 2y + 4xy = 2y(2x + 1) \\ v(x, y) &= \frac{\partial}{\partial x} \left(y^2 + 2xy^2 - \frac{x^3}{3} - \frac{x^2}{2} \right) \implies v(x, y) = x^2 + x - 2y^2 = x(x + 1) - 2y^2. \end{aligned}$$

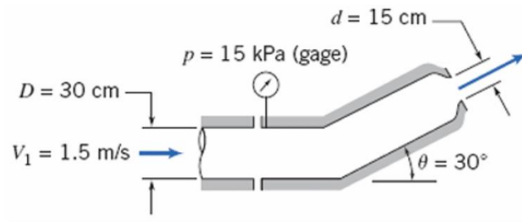
Which means the stream function is correct.

$$\psi(x, y) = y^2 + 2xy^2 - \frac{x^3}{3} - \frac{x^2}{2}$$

RESULT

Problem 4

Water flows steadily through the nozzle, shown in the figure, discharging to atmosphere. Calculate the horizontal component of force in the flanged joint. Indicate whether the joint is in tension or compression. Use density of water = $1000 \frac{\text{kg}}{\text{m}^3}$.



Assumptions

- Steady flow
- Incompressible flow
- Uniform flow
- Negligible mass (negligible body forces)

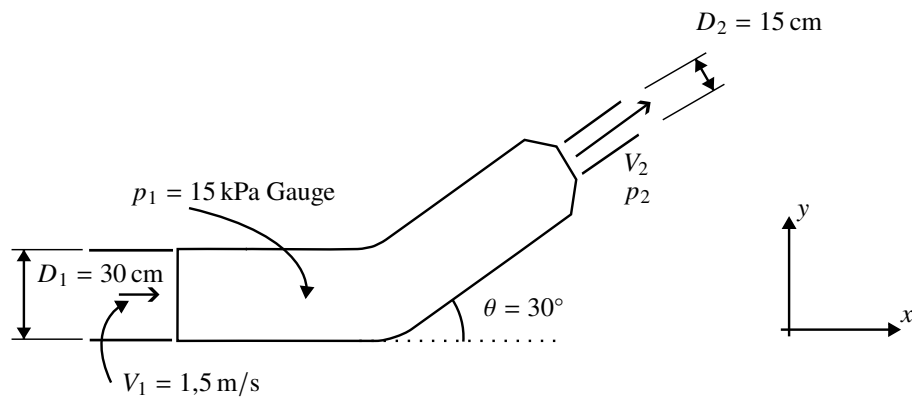


Figure 4.1: Sketch of the joint.

Derivation

We can once again apply the momentum equation for an inertial control volume, [Equation \(1.1\)](#),

$$\mathbf{F} = \mathbf{F}_B + \mathbf{F}_S = \frac{\partial}{\partial t} \int_{CV} \mathbf{V} \rho \, dV + \int_{CS} \mathbf{V} \rho \mathbf{V} \cdot d\mathbf{A}.$$

Following the same argumentation as in [Problem 1](#) with the same assumptions this reduces to the sum from [Equation \(1.2\)](#)

$$F_x = F_{S_x} = \sum_{CS} u \rho \mathbf{V} \cdot \mathbf{A}.$$

Hence,

$$R_x + p_1 \cdot A_1 + p_2 \cdot A_2 = V_1 \cdot (-\rho \cdot V_1 \cdot A_1) + V_2 \cdot \cos \theta \cdot (\rho \cdot V_2 \cdot A_2) \quad (4.1a)$$

$$R_x = -p_1 \cdot A_1 + \rho \left(V_2^2 \cdot \cos \theta \cdot A_2 - V_1^2 A_1 \right). \quad (4.1b)$$

From the continuity equation we have that

$$A_2 V_2 = A_1 V_1$$

$$\begin{aligned}
 V_2 &= V_1 \frac{A_1}{A_2} \\
 &= V_1 \left(\frac{D_1}{D_2} \right)^2 \\
 &= 1,5 \frac{\text{m}}{\text{s}} \cdot \left(\frac{30 \text{ cm}}{15 \text{ cm}} \right)^2 \\
 &= 6 \frac{\text{m}}{\text{s}}.
 \end{aligned}$$

Now plugging all the known values into [Equation \(4.1b\)](#) we obtain

$$\begin{aligned}
 R_x &= -15 \text{ kPa} \cdot \pi \cdot \frac{(30 \text{ cm})^2}{4} + 1000 \frac{\text{kg}}{\text{m}^3} \cdot \left(\left(6 \frac{\text{m}}{\text{s}} \right)^2 \cdot \cos 30^\circ \cdot \pi \cdot \frac{(15 \text{ cm})^2}{4} - \left(1,5 \frac{\text{m}}{\text{s}} \right)^2 \cdot \pi \cdot \frac{(30 \text{ cm})^2}{4} \right) \\
 &= -668,39 \text{ N}.
 \end{aligned}$$

Therefore, the reaction force in the joint will be in the $-x$ -direction, as defined on [Figure 4.1](#), and the joint will be in tension. This aligns with the expectation given that it needs to withstand high pressure and a large change in momentum of the fluid.

$R_x = 668 \text{ N(Tension)}$

RESULT

Problem 5

The velocity components for an incompressible steady flow field are given below

$$u = A (x^2 + z^2) \quad v = B (xy + yz).$$

Determine the z -component of velocity for steady and unsteady flow.

Assumptions

- Steady flow
- Incompressible flow

Derivation

The continuity equation on differential form can be written as

$$\nabla \cdot \rho \mathbf{V} + \frac{\partial \rho}{\partial t} = 0 \quad (5.1)$$

As we are assuming the flow to be incompressible $\rho = \text{const.}$ This means that $\frac{\partial \rho}{\partial t} = 0$ and $\nabla \cdot \rho \mathbf{V} = \nabla \cdot \mathbf{V}$ in [Equation \(5.1\)](#) which therefore reduces to

$$\nabla \cdot \mathbf{V} = 0 \quad (5.2)$$

for an incompressible flow. On Cartesian form [Equation \(5.2\)](#) becomes

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0.$$

Substituting in our known values we get

$$\begin{aligned} \frac{\partial}{\partial x} (A (x^2 + z^2)) + \frac{\partial}{\partial y} (B (xy + yz)) + \frac{\partial w}{\partial z} &= 0 \\ -\frac{\partial w}{\partial z} &= 2Ax + B (x + z) \\ \frac{\partial w}{\partial z} &= -2Ax - Bx - Bz \\ w &= -\int (2A + B) x \, dz - \int Bz \, dz \\ &= -(2A + B) xz - \frac{B}{2} z^2 + f(x, y) \end{aligned}$$

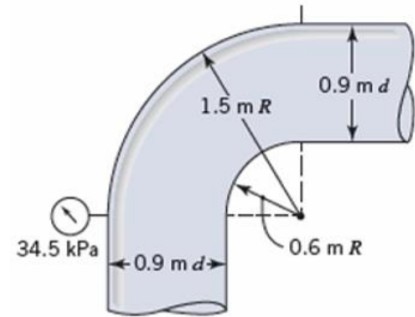
where $f(x, y)$ is a general function of x and y . As the time-variant term $\frac{\partial \rho}{\partial t} = 0$ in [Equation \(5.1\)](#) for both steady and unsteady flows, as long as they are incompressible as in this case $\rho = \text{const.}$ Therefore the general expression for the z -component of the velocity will be the same for both steady and unsteady flow.

$$w = -(2A + B)xz - \frac{B}{2}z^2 + f(x, y)$$

RESULT

Problem 6

The water flow rate through the vertical bend shown in the figure below is $2,83 \frac{\text{m}^3}{\text{s}}$. Calculate the magnitude, direction, and location of the resultant force of the water pipe bend. Use density of water = $999 \frac{\text{kg}}{\text{m}^3}$.



Assumptions

- Steady flow
- Incompressible flow
- Inviscid flow

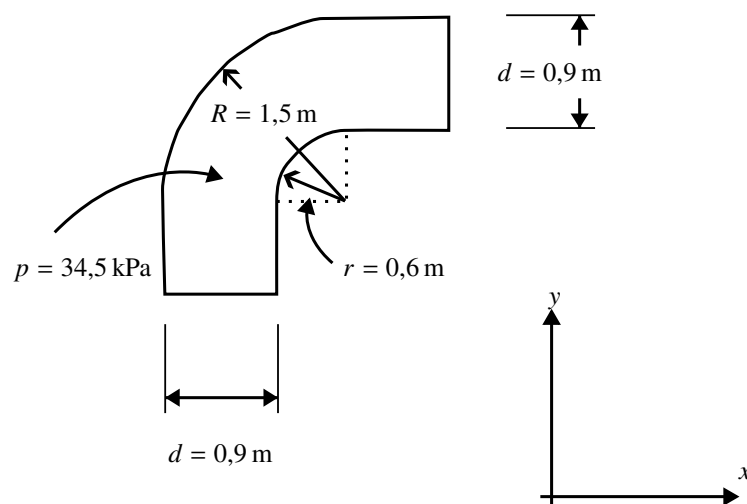


Figure 6.1: Sketch of the vertical bend.

Derivation

The continuity equation is generally

$$\frac{\partial}{\partial t} \int_{CV} \rho dV + \int_{CS} \rho \mathbf{V} \cdot d\mathbf{A} = 0. \quad (6.1)$$

For a steady incompressible flow Equation (6.1) reduces to Equation (1.5a),

$$Q = A_n V_n = \text{const.}$$

For any point n . Using this we can calculate the flow velocity through the bend, as

$$V = \frac{Q}{A} = \frac{Q}{\pi \cdot \frac{d^2}{4}} = \frac{2,83 \frac{\text{m}^3}{\text{s}}}{\pi \cdot \frac{(0,9 \text{ m})^2}{4}} = 4,45 \frac{\text{m}}{\text{s}}.$$

As shown in Equation (1.5a) flow speed V will be constant for constant area $A = A_1 = A_2$, hence the flow speed through the entire bend must be $V = V_1 = V_2 = 4,45 \frac{\text{m}}{\text{s}}$.

The Bernoulli equation is

$$\frac{p}{\rho} + \frac{V^2}{2} + gh = \text{constant}. \quad (6.2)$$

Using the Bernoulli equation, Equation (6.2), we can find the pressure p_2 at the outlet of the bend as

$$\begin{aligned} \frac{p_1}{\rho} + \frac{V^2}{2} + g \cdot 0 &= \frac{p_2}{\rho} + \frac{V^2}{2} + gh \\ \frac{p_2}{\rho} &= \frac{p_1}{\rho} - gh \\ p_2 &= p_1 - \rho gh \\ &= p_1 - \rho g \left(r + \frac{d}{2} \right) \\ &= 34,5 \text{ kPa} - 999 \frac{\text{kg}}{\text{m}^3} \cdot 9,81 \frac{\text{m}}{\text{s}^2} \cdot \left(0,6 \text{ m} + \frac{0,9 \text{ m}}{2} \right) \\ &= 24,2 \text{ kPa}. \end{aligned}$$

Note that in the above the height difference h has been calculated as the median height difference for the water. Now we can use the momentum equation for an inertial control volume, Equation (1.1),

$$\mathbf{F} = \mathbf{F}_B + \mathbf{F}_S = \frac{\partial}{\partial t} \int_{CV} \mathbf{V} \rho \, dV + \int_{CS} \mathbf{V} \rho \mathbf{V} \cdot d\mathbf{A}.$$

In the case of steady flow Equation (1.1) reduces to

$$\mathbf{F}_B + \mathbf{F}_S = \int_{CS} \mathbf{V} \rho \mathbf{V} \cdot d\mathbf{A}. \quad (6.3)$$

Splitting Equation (6.3) into Cartesian coordinates yields

$$(F_B)_x + (F_S)_x = \int_{CS} u \rho \mathbf{V} \cdot d\mathbf{A} \quad (6.4a)$$

$$(F_B)_y + (F_S)_y = \int_{CS} v \rho \mathbf{V} \cdot d\mathbf{A}. \quad (6.4b)$$

As $V = V_1 = V_2 = 4,45 \frac{\text{m}}{\text{s}}$, $u = v = V = 4,45 \frac{\text{m}}{\text{s}}$ in Equations 6.4. From Equation (6.4a) we get

$$\begin{aligned} R_x + (F_B)_x + (F_S)_x &= V^2 \cdot \rho \cdot A \\ R_x - p_2 A &= V^2 \cdot \rho \cdot A \\ R_x &= \pi \cdot \frac{d^2}{4} \left(V^2 \cdot \rho + p_2 \right) \\ &= \pi \cdot \frac{(0,9 \text{ m})^2}{4} \left(\left(4,45 \frac{\text{m}}{\text{s}} \right)^2 \cdot 999 \frac{\text{kg}}{\text{m}^3} + 24,2 \text{ kPa} \right) \\ &= 28,0 \text{ kN}. \end{aligned}$$

I.e. the water will experience a reaction force of $R_x = 28,0 \text{ kN}$ from the bend, which means that the water will exert a force $F_x = -R_x = -28,0 \text{ kN}$ on the bend.

From Equation (6.4b) we get

$$R_y + (F_B)_y + (F_S)_y = -V^2 \cdot \rho \cdot A$$

$$\begin{aligned}
R_y + \left(\rho g A \frac{2\pi}{4} h \right) + (p_1 A) &= -V^2 \cdot \rho \cdot A \\
R_y &= -V^2 \cdot \rho \cdot A - p_1 A - \rho g A \frac{2\pi}{4} h \\
&= -A \left(p_1 + \rho \left(V^2 + g \frac{\pi}{2} h \right) \right) \\
&= -\pi \cdot \frac{d^2}{4} \left(p_1 + \rho \left(V^2 + g \frac{\pi}{2} \left(r + \frac{d}{2} \right) \right) \right) \\
&= -\pi \cdot \frac{(0,9 \text{ m})^2}{4} \left(34,5 \text{ kPa} + 999 \frac{\text{kg}}{\text{m}^3} \left(\left(4,45 \frac{\text{m}}{\text{s}} \right)^2 + \left(-9,81 \frac{\text{m}}{\text{s}^2} \right) \frac{\pi}{2} \left(0,6 \text{ m} + \frac{0,9 \text{ m}}{2} \right) \right) \right) \\
&= -24,2 \text{ kN}.
\end{aligned}$$

This means the water will experience a reaction force of $R_y = -24,2 \text{ kN}$ from the bend, hence, the water will exert a force $F_y = -R_y = 24,2 \text{ kN}$ on the bend. Using F_x and F_y the total force exerted by the water on the bend can be computed as

$$F = \sqrt{F_x^2 + F_y^2} = \sqrt{(-28,0 \text{ kN})^2 + (24,2 \text{ kN})^2} = 37,0 \text{ kN}.$$

The direction of the force can be found using trigonometry as

$$\begin{aligned}
\tan \alpha &= \frac{F_y}{F_x} \\
\alpha &= \tan^{-1} \left(\frac{F_y}{F_x} \right) \\
&= \tan^{-1} \left(\frac{24,2 \text{ kN}}{-28 \text{ kN}} \right) \\
&= 139,16^\circ.
\end{aligned}$$

Hence the water will exert a force with a size of $F = 37,0 \text{ kN}$ at an angle of 139° with respect to the $+x$ direction.

$F = 37,0 \text{ kN} \angle 139^\circ$

RESULT

Problem 7

A viscous liquid is sheared between two parallel disks of radius R , one of which rotates while the other is fixed. The velocity field is purely tangential, and the velocity varies linearly with z from $V_\phi = 0$ at $z = 0$ (the fixed disk) to the velocity of the rotating disk at its surface ($z = h$).

Derive an expression for the velocity field between the disks.

Assumptions

- Incompressible flow
- Purely tangential flow
- Velocity varies linearly with z

Derivation

Here we will apply the continuity equation on cylindrical form

$$\frac{1}{r} \frac{\partial}{\partial r} r \rho V_r + \frac{1}{r} \frac{\partial}{\partial \phi} \rho V_\phi + \frac{\partial}{\partial z} \rho V_z + \frac{\partial \rho}{\partial t} = 0 \quad (7.1)$$

As the flow is assumed incompressible $\frac{\partial \rho}{\partial t} = 0$ in Equation (7.1) and the assumption of purely tangential flow means $V_r = V_z = 0$ in Equation (7.1). Hence the continuity equation reduces to

$$\frac{\partial V_\phi}{\partial \phi} = 0.$$

This means that the velocity V_ϕ is not a function of ϕ and hence $V_\phi = V_\phi(r, z)$. As we know the velocity scales linearly with z we can write the function for z as a normal linear function,

$$V_\phi(r, z) = z \cdot f(r) + C \quad (7.2)$$

Applying the given boundary conditions of $V_\phi(r, 0) = 0$ and $V_\phi(r, h) = r\omega$ to Equation (7.2) yields

$$V_\phi(r, 0) = 0 \quad (7.3a)$$

$$0 \cdot f(r) + C = 0 \quad (7.3b)$$

$$\implies C = 0 \quad (7.3c)$$

$$V_\phi(r, h) = r\omega \quad (7.3d)$$

$$\implies h \cdot f(r) = r\omega \quad (7.3e)$$

$$\implies f(r) = \frac{r\omega}{h}. \quad (7.3f)$$

Substituting Equation (7.3f) into Equation (7.2) gives an expression for the tangential velocity V_ϕ

$$V_\phi = \frac{r\omega}{h} z. \quad (7.4)$$

The velocity flow field on cylindrical coordinates is

$$\mathbf{V} = V_r \hat{e}_r + V_\phi \hat{e}_\phi + V_z \hat{k} \quad (7.5)$$

In Equation (7.5) $V_r = V_z = 0$ since the flow is assumed purely tangential. Therefore the entire velocity flow field is given by substituting Equation (7.4) into Equation (7.5),

$$\mathbf{V} = V_\phi \hat{e}_\phi = \frac{r\omega}{h} z \hat{e}_\phi.$$

$$\mathbf{V} = \frac{r\omega}{h} z \hat{e}_\phi$$

RESULT